VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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CERTIFICATE

B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering

This is to certify that the Lab work entitled **"ANALYSIS AND DESIGN OF ALGORITHMS"** carried out by ARPITA GANGADHAR PATTAR **(1BM24CS053)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Index Sheet

Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1: Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>


void topologicalSort(int n, int adj[100][100])
{
   int queue[100];
   int indegree[100];
   int front = 0, rear = -1;
   for (int i = 0; i < n; i++)
   {
      indegree[i] = 0;
   }
   for (int i = 0; i < n; i++)
   {
      for (int j = 0; j < n; j++)
      {
         if (adj[i][j] == 1)
         {
            indegree[j]++;
         }
      }
   }
   for (int i = 0; i < n; i++)
   {
      if (indegree[i] == 0)
      {
         queue[++rear] = i;
      }
   }
```

```
    printf("Topological Order: ");
    while (front <= rear)
    {
        int node = queue[front++];
        printf("%d ", node);

        for (int i = 0; i < n; i++)
        {
            if (adj[node][i] == 1)
            {
                indegree[i]--;

                if (indegree[i] == 0)
                {
                    queue[++rear] = i;
                }
            }
        }
    }
    printf("\n");
}
int main()
{
    int n;
    int adj[100][100];

    printf("Enter the number of vertices: ");
    scanf("%d", &n);
```

```
    printf("Enter the adjacency matrix:\n");

    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            scanf("%d", &adj[i][j]);
        }
    }

    topologicalSort(n, adj);

    return 0;
}
```

OUTPUT:

```
Enter the number of vertices: 5
Enter the adjacency matrix:
0 1 1 0 0
0 0 0 1 1
0 1 0 1 0
0 0 0 0 0
0 0 0 0 0
Topological Order: 0 2 1 3 4

Process returned 0 (0x0)   execution time : 29.539 s
Press any key to continue.
```

Lab Program 2: Implement Johnson Trotter algorithm to generate permutations.

```
#include <stdio.h>
int LR = 1, RL = 0;


void swap(int *a, int *b)
{
   int temp = *a;
   *a = *b;
   *b = temp;
}
int fact(int n)
{
   int result = 1;
   for (int i = 1; i <= n; i++)
      result = result * i;
   return result;
}


int permutation(int a[], int dir[], int n)
{
   int prev = 0, mobile = 0, pos = 0;
   for (int i = 0; i < n; i++)
   {
      if (i != 0 && dir[a[i] - 1] == RL)
      {
         if (a[i] > a[i - 1] && a[i] > prev)
         {
```

```
            mobile = a[i];
            prev = mobile;
        }
    }
    if (i != n - 1 && dir[a[i] - 1] == LR)
    {
        if (a[i] > a[i + 1] && a[i] > prev)
        {
            mobile = a[i];
            prev = mobile;
        }
    }
}
for (int i = 0; i < n; i++)
{
    if (a[i] == mobile)
        pos = i + 1;
}
if (dir[a[pos - 1] - 1] == RL)
    swap(&a[pos - 1], &a[pos - 2]);
else if (dir[a[pos - 1] - 1] == LR)
    swap(&a[pos], &a[pos - 1]);

for (int i = 0; i < n; i++)
{
    if (a[i] > mobile)
    {
```

```
            if (dir[a[i] - 1] == LR)
                dir[a[i] - 1] = RL;
            else if (dir[a[i] - 1] == RL)
                dir[a[i] - 1] = LR;
        }
    }

    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
    return 0;
}

int main()
{
    int n;
    printf("Enter the number of n: ");
    scanf("%d", &n);
    int a[n], dir[n];

    for (int i = 0; i < n; i++)
    {
        a[i] = i + 1;
        printf("%d ", a[i]);
    }
    printf("\n");
```

```
    for (int i = 0; i < n; i++)
        dir[i] = RL;
    for (int i = 1; i < fact(n); i++)
        permutation(a, dir, n);

    return 0;
}
```

Ouput:

Lab Program 3: Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include<stdio.h>


void MergeSort(int a[],int b[],int m,int n)
{
   int i=0,j=0,k=0,c[100];
   while(i<m && j<n)
   {
      if(a[i]<b[j])
       {
          c[k]=a[i];
         k++;
         i++;
       }
      else
      {
         c[k]=b[j];
         k++;
         j++ ;
      }
   }
   while(i<m)
   {
      c[k]=a[i];
      k++;
      i++;
```

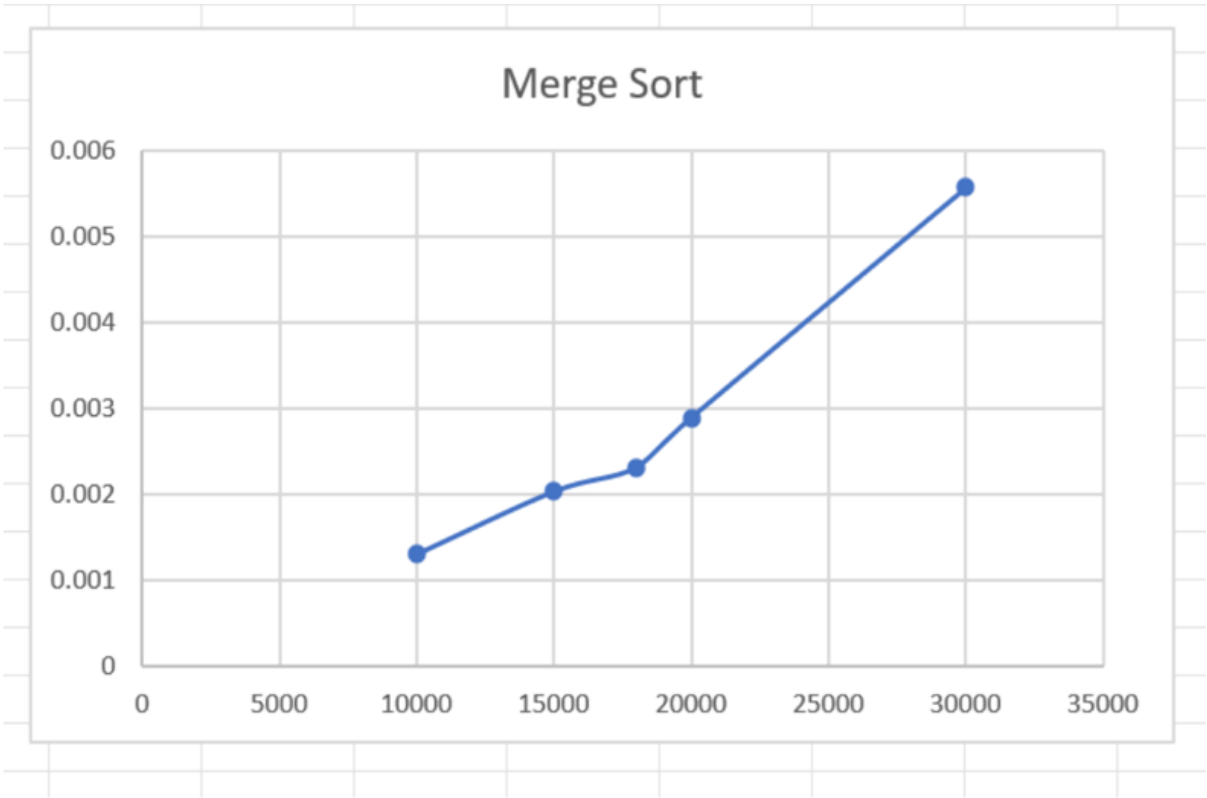
```
    }
    while(j<n)
    {
        c[k]=b[j];
        k++;
        j++;
    }
    printf("Merged Array:");
    for(int i=0;i<(m+n);i++)
        printf("%d\t",c[i]);
}

int main()
{
    int n,m,a[100],b[100];
    printf("Enter the number of elements in First array:");
    scanf("%d",&m);
    printf("Enter the number of elements of first array:");
    for(int i=0;i<m;i++)
        scanf("%d",&a[i]);

    printf("Enter the number of elements in Second array:");
    scanf("%d",&n);
    printf("Enter the number of elements of first array:");
    for(int i=0;i<n;i++)
        scanf("%d",&b[i]);
```

```
    MergeSort(a,b,m,n);
    return 0;
}
```

Output:

```
Enter the number of elements in First array:5
Enter the number of elements of first array:1 3 5 7 9
Enter the number of elements in Second array:5
Enter the number of elements of first array:2 4 6 8 10
Merged Array:1  2       3       4       5       6       7       8       9       10
Process returned 0 (0x0)   execution time : 15.473 s
Press any key to continue.
```

Graph:



Lab Program 4: Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

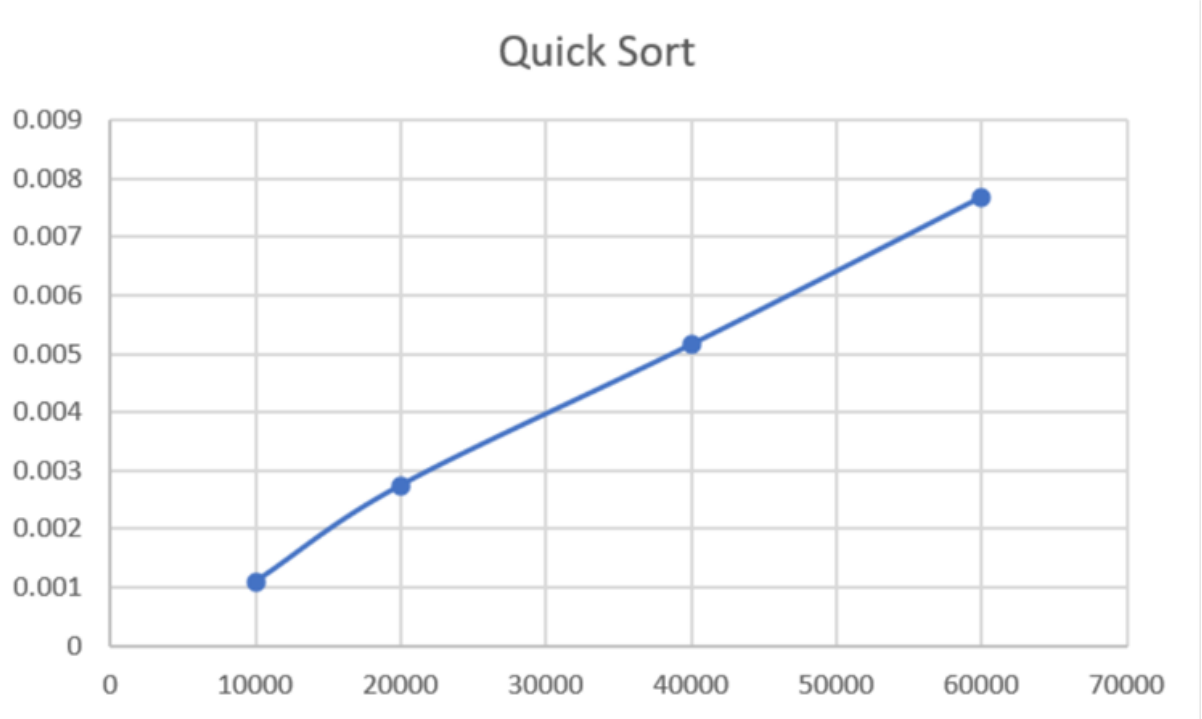
```
#include<stdio.h>


int partition(int a[] ,int low,int high)
{
   int i,j,temp,key;
   i=low+1;
   j=high;
   key=a[low];
   while(1)
   {
      while(a[i]<=key && i<=high)
         i++;
      while(a[j]>key)
         j--;
      if(i<j){
         temp=a[i];
         a[i]=a[j];
         a[j]=temp;
      }
      else{
         temp=a[low];
         a[low]=a[j];
         a[j]=temp;
         return j;
      }
   }
}
```

```
void QuickSort(int a[],int low,int high)
{
   int j;
   if(low<high)
   {
       j=partition(a,low,high);
       QuickSort(a,low,j-1);
       QuickSort(a,j+1,high);
   }

}

int main()
{
   int n,a[100];
   printf("Enter the number of elements in the array:");
   scanf("%d",&n);
   printf("Enter the array elements:");
   for(int i=0;i<n;i++)
      scanf("%d",&a[i]);
   QuickSort(a,0,n-1);

   printf("Sorted Array:");
   for(int i=0;i<n;i++)
      printf("%d\t",a[i]);

   return 0;

}
```

Output:

```
Enter the number of elements in the array:6
Enter the array elements:12 5 8 23 56 89
Sorted Array:5  8       12      23      56      89
Process returned 0 (0x0)   execution time : 11.668 s
Press any key to continue.
```

Graph:



Lab Program 5: Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include<stdio.h>


void heapify(int a[],int n,int p)
{
   int c,item;
   item=a[p];
   c=2*p+1;


   while(c<n)
   {
      if(c+1<n)
      {
         if(a[c]<a[c+1])
            c++;
      }
      if(item<a[c])
      {
         a[p]=a[c];
         p=c;
         c=2*p+1;
      }
      else
         break;


   }
   a[p]=item;
}
```

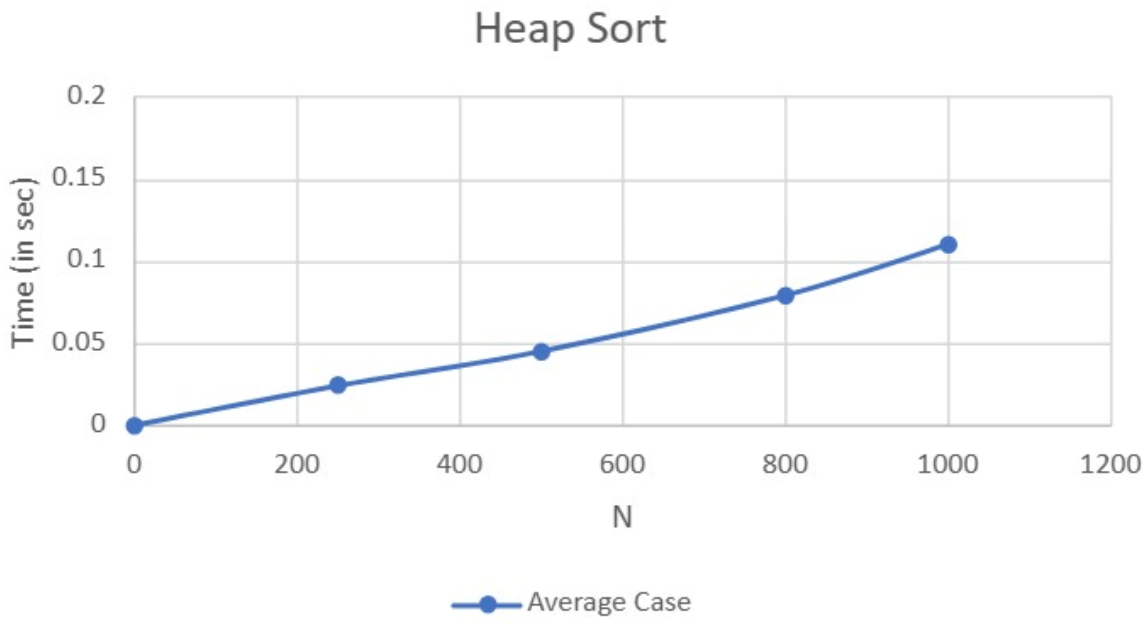
```
void Build_heap(int a[],int n)
{
    for(int i=(n-1)/2;i>=0;i--)
        heapify(a,n,i);
}

void heapsort(int a[],int n)
{
    int i,temp;
    Build_heap(a,n);

    for(int i=n-1;i>0;i--)
    {
        temp=a[0];
        a[0]=a[i];
        a[i]=temp;
        heapify(a,i,0);
    }
    printf("Sorted Array:");
    for(int i=0;i<n;i++)
        printf("%d\t",a[i]);
}
int main()
{
    int a[100],n,i;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    printf("Enter the array elements: ");
    for(i=0;i<n;i++)
```

```
        scanf("%d",&a[i]);
    heapsort(a,n);
    return 0;
}
```

Output:

```
Enter the number of elements:8
Enter the array elements: 13 8 5 2 1 19 26 15
Sorted Array:1  2       5       8       13      15      19      26
Process returned 0 (0x0)   execution time : 24.709 s
Press any key to continue.
```

Graph:

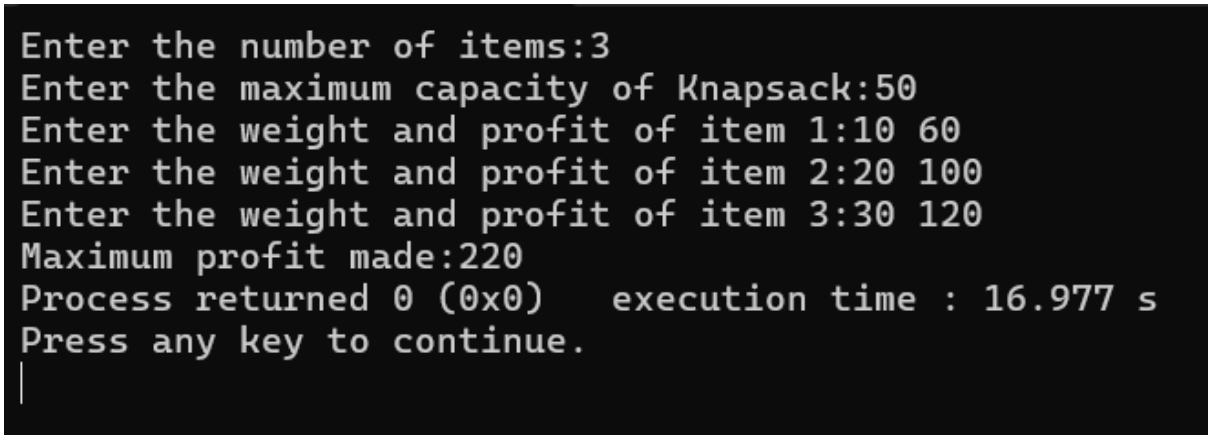


Lab Program 6: Implement 0/1 Knapsack problem using dynamic programming.

```
#include<stdio.h>

int max(int a,int b)
{
   if(a>b)
      return a;
   return b;
}

int main()
{
   int p[100],w[100],M,n,table[100][100];
   printf("Enter the number of items:");
   scanf("%d",&n);
   printf("Enter the maximum capacity of Knapsack:");
   scanf("%d",&M);
   for(int i=1;i<=n;i++){
      printf("Enter the weight and profit of item %d:",i);
      scanf("%d %d",&w[i],&p[i]);
   }
   for(int i=0;i<=n;i++){
      for(int j=0;j<=M;j++)
         if(i==0 || j==0) {
            table[i][j]=0;
         }
         else if(j<w[i]) {
            table[i][j]=table[i-1][j];
         }
```

```
        else {
            table[i][j]=max(table[i-1][j],p[i]+table[i-1][j-w[i]]);
        }
    }
    printf("Maximum profit made:%d",table[n][M]);
    return 0;
}
```

Output:



Lab Program 7: Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include<stdio.h>
#define INF 99

void Floyds(int a[10][10],int n)
{   for(int k=0;k<n;k++)
    {
      for(int i=0;i<n;i++)
      {
        for(int j=0;j<n;j++)
           if(a[i][k]+a[k][j]<a[i][j])
              a[i][j]=a[i][k]+a[k][j];
      }
    }
    printf("Shortest Distance Matrix:\n");
    for(int i=0;i<n;i++)
    {
       for(int j=0;j<n;j++)
           if(a[i][j]==INF)
              printf("INF");
           else
              printf("%d ",a[i][j]);
      printf("\n");
    }
}

int main()
{
    int i,j,k,n;
```

```
    int cost[10][10],a[10][10];
    printf("Enter the number of Vertices:");
    scanf("%d",&n);
    printf("Enter the cost matrix:\n");
    for(i=0;i<n;i++)
    {
       for(j=0;j<n;j++)
       {
          scanf("%d",&cost[i][j]);
          a[i][j]=cost[i][j];
       }
    }
    Floyds(a,n);
    return 0;
}
```

Output:

```
Enter the number of Vertices:5
Enter the cost matrix:
0 4 99 5 99
99 0 1 99 6
2 99 0 3 99
99 99 1 0 2
1 99 99 4 0
Shortest Distance Matrix:
0 4 5 5 7
3 0 1 4 6
2 6 0 3 5
3 7 1 0 2
1 5 5 4 0

Process returned 0 (0x0)   execution time : 36.980 s
Press any key to continue.
```

Lab Program 8:(a) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>
#include <time.h>

#define MAX INT_MAX

int minKey(int key[], int set[], int n)
{
   int min = MAX, min_id;
   for (int i = 0; i < n; i++)
   {
      if (set[i] == 0 && key[i] < min)
      {
         min = key[i];
         min_id = i;
      }
   }
   return min_id;
}
int main()
{
   int n, parent[100], key[100], set[100], graph[100][100];
   int mincost = 0;
   clock_t start, end;
   printf("Enter the number of vertices in the graph: ");
   scanf("%d", &n);
   printf("Enter the adjacency matrix:\n");
   for (int i = 0; i < n; i++)
   {
      for (int j = 0; j < n; j++)
      {
```

```
            scanf("%d", &graph[i][j]);
        }
    }
    start = clock();

    for (int i = 0; i < n; i++)
    {
        key[i] = MAX;
        set[i] = 0;
    }
    key[0] = 0;
    parent[0] = -1;

    for (int count = 0; count < n - 1; count++)
    {
        int i = minKey(key, set, n);
        set[i] = 1;
        for (int j = 0; j < n; j++)
        {
            if (graph[i][j] &&
                set[j] == 0 &&
                graph[i][j] < key[j])
            {
                parent[j] = i;
                key[j] = graph[i][j];
            }
        }
    end = clock();
    printf("\nEdge\tWeight\n");
    for (int i = 1; i < n; i++)
    {
        printf("%d - %d\t%d\n",parent[i],i, graph[parent[i]][i]);
```

```
        mincost += graph[parent[i]][i];
    }
    printf("\nMinimum cost is %d\n", mincost);
    double time = (double)(end - start) / CLOCKS_PER_SEC;
    printf("Execution time: %f\n", time);
    return 0;
}
```

Output:

```
Enter the number of vertices in the graph: 5
Enter the adjacency matrix:
0 2 0 6 0
2 0 3 8 5
0 3 0 0 7
6 8 0 0 9
0 5 7 9 0

Edge    Weight
0 - 1   2
1 - 2   3
0 - 3   6
1 - 4   5

Minimum cost is 16
Execution time: 0.000000

Process returned 0 (0x0)   execution time : 50.840 s
Press any key to continue.
```

Lab Program 8:(b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include<stdio.h>

typedef structs
{
  int u,v,w;
}Edge;

Edge edges[100],result[100];
int parent[100];

int find(int i)
{
   while(parent[i]!=i)
   {
      i=parent[i];
   }
   return i;
}

void unionSet(int i,int j)
{
   parent[i]=j;
}
void sortEdge(int e)
{
   int i,j;
   Edge temp;
   for(i=0;i<e-1;i++){
      for(j=0;j<e-1-i;j++)
      {
```

```
            if(edges[j].w>edges[j+1].w){
                temp=edges[j];
                edges[j]=edges[j+1];
                edges[j+1]=temp;
            }
        }
    }
}

int kruskals(int n,int e)
{
    int count=0;
    int cost=0;
    int u,v;
    for(int i=0;i<n;i++){
        parent[i]=i;
    }
    sortEdge(e);
    for(int i=0;i<e&& count<n-1;i++)
    {
        u=find(edges[i].u);
        v=find(edges[i].v);
        if(u!=v){
            result[count++]=edges[i];
            cost+=edges[i].w;
            unionSet(u,v);
        }
    }
    printf("Minimum Spanning Tree:\n");
    for(int i=0;i<count;i++){
        printf("%d---%d = %d\n",result[i].u,result[i].v,result[i].w);
    }
```

```
    return cost;
}
```

Output:

```
Enter the number of vertices:7
Enter the number of edges:9
Enter the edges(u,v,weight):
1 2 4
1 3 28
2 5 26
3 6 12
3 4 8
4 5 22
4 7 16
6 7 6
5 7 20
Minimum Spanning Tree:
1---2 = 4
6---7 = 6
3---4 = 8
3---6 = 12
5---7 = 20
2---5 = 26
Total Cost:76

Process returned 0 (0x0)   execution time : 55.094 s
Press any key to continue.
```

Lab Program 9: Implement Fractional Knapsack using Greedy technique.

```
#include <stdio.h>

int main() {
    int i, j, n, m;
    int p[50], w[50];
    float x[50];
    int t1, t2;

    printf("Enter the number of objects: ");
    scanf("%d", &n);
    printf("Enter the weights of all objects:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &w[i]);
    }
    printf("Enter the profit of all objects:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &p[i]);
    }
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if ((float)p[i] / w[i] < (float)p[j] / w[j]) {

                t1 = p[i];
                p[i] = p[j];
                p[j] = t1;



                t2 = w[i];
```

```
            w[i] = w[j];
            w[j] = t2;
        }
    }
}
printf("Enter the maximum capacity: ");
scanf("%d", &m);

for (i = 0; i < n; i++) {
    x[i] = 0.0;
}
int rem_cap = m;
for (i = 0; i < n; i++) {
    if (w[i] > rem_cap) {
        break;
    }
    x[i] = 1.0;
    rem_cap -= w[i];
}
if (i < n) {
    x[i] = (float)rem_cap / w[i];
}
float sum = 0;
for (i = 0; i < n; i++) {
    sum = sum + (p[i] * x[i]);
}
printf("Total profit : %f\n", sum);
return 0;
}
```

Output:

```
Enter the number of objects: 3
Enter the weights of all objects:
18 15 10
Enter the profit of all objects:
25 24 15
Enter the maximum capacity: 20
Total profit : 31.500000

Process returned 0 (0x0)   execution time : 13.103 s
Press any key to continue.
```

Lab Program 10: From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm

```
#include <stdio.h>
#include <time.h>

#define n 5

int minDist(int dist[], int visited[]) {
   int min = 99999, min_index = -1;
   for (int i = 0; i < n; i++) {
      if (visited[i] == 0 && dist[i] <= min) {
         min = dist[i];
         min_index = i;
      }
   }
   return min_index;
}

int dijkstra(int weight[n][n], int src) {
   int dist[n];
   int visited[n];
   for (int i = 0; i < n; i++) {
      dist[i] = 99999;
      visited[i] = 0;
   }
   dist[src] = 0;
   for (int count = 0; count < n - 1; count++) {
      int u = minDist(dist, visited);
      visited[u] = 1;
      for (int v = 0; v < n; v++) {
```

```
            if (!visited[v] && weight[u][v] != 0 && dist[u] != 99999) {
                int new_dist = dist[u] + weight[u][v];
                if (new_dist < dist[v]) {
                    dist[v] = new_dist;
                }
            }
        }
    }
    printf("Vertex \t Distance from source\n");
    for (int i = 0; i < n; i++) {
        printf("%d \t\t %d\n", i + 1, dist[i]);
    }
    return 0;
}
int main() {
    int i, j;
    int graph[n][n];
    clock_t start, end;

    printf("Enter the weight matrix (%dx%d):\n", n, n);
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &graph[i][j]);
        }
    }
    int source = 0;
    start = clock();
    dijkstra(graph, source);
    end = clock();
```

```
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
    printf("Time taken: %f seconds\n", time_taken);

    return 0;
}
```

Output:

```
Enter the weight matrix (5x5):
0 10 0 30 100
10 0 50 0 0
0 50 0 20 10
30 0 20 0 60
100 0 10 60 0
Vertex   Distance from source
1                0
2                10
3                50
4                30
5                60
Time taken: 0.001000 seconds

Process returned 0 (0x0)   execution time : 124.393 s
Press any key to continue.
```

Lab Program 11: Implement "N-Queens Problem" using Backtracking.

```
#include <stdio.h>
#include <stdbool.h>

int solutionCount = 0;

bool isSafe(int N, int board[N][N], int row, int col) {
    int i, j;
    for (i = 0; i < col; i++) {
        if (board[row][i]) {
            return false;
        }
    }
    for (i = row, j = col; i >= 0 && j >= 0; i--, j--) {
        if (board[i][j]) {
            return false;
        }
    }
    for (i = row, j = col; j >= 0 && i < N; i++, j--) {
        if (board[i][j]) {
            return false;
        }
    }
    return true;
}
void printSolution(int N, int board[N][N]) {
    printf("Solution %d:\n", ++solutionCount);
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
```

```
            printf("%d ", board[i][j]);
        }
        printf("\n");
    }
    printf("\n");
}
void solveNQueenRecursive(int N, int board[N][N], int col) {
    if (col >= N) {
        printSolution(N, board);
        return;
    }
    for (int i = 0; i < N; i++) {
        if (isSafe(N, board, i, col)) {
            board[i][col] = 1;
            solveNQueenRecursive(N, board, col + 1);
            board[i][col] = 0;
        }
    }
}
void solveNQ(int N) {
    int board[N][N];
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            board[i][j] = 0;
        }
    }
    solveNQueenRecursive(N, board, 0);
    if (solutionCount == 0) {
        printf("Solution does not exist\n");
```

```
    } else {
        printf("Total solutions = %d\n", solutionCount);
    }
}
int main() {
    int N;
    printf("Enter value of N: ");
    scanf("%d", &N);
    if (N <= 0) {
        printf("Enter a positive number.\n");
        return 0;
    }
    solveNQ(N);
    return 0;
}
```

Output:

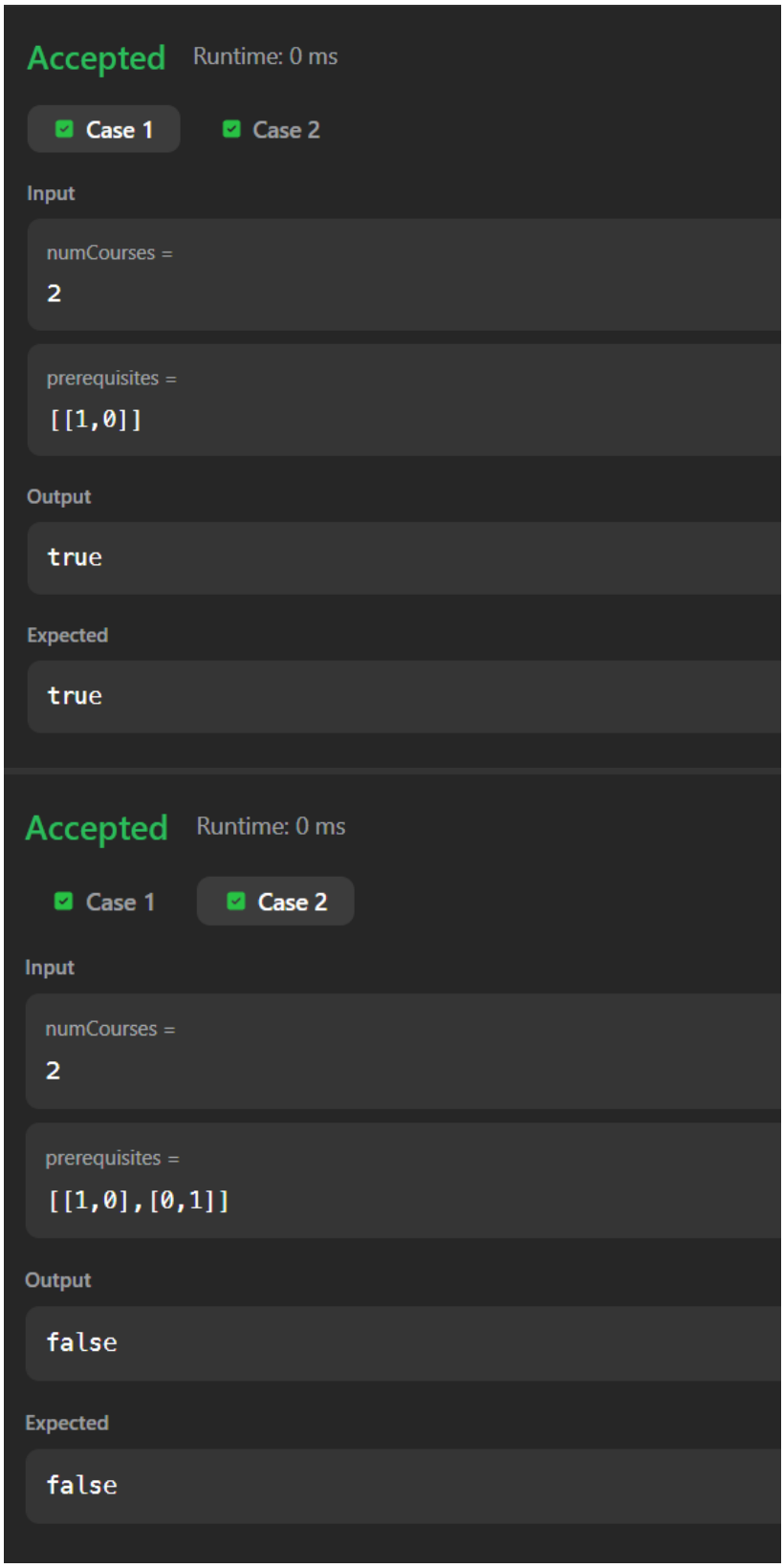
```
Enter value of N: 4
Solution 1:
0 0 1 0
1 0 0 0
0 0 0 1
0 1 0 0

Solution 2:
0 1 0 0
0 0 0 1
1 0 0 0
0 0 1 0

Total solutions = 2

Process returned 0 (0x0)   execution time : 14.530 s
Press any key to continue.
```

Leetcode: 1)Leet Code exercises on topological sorting(207).

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {
    int inDegree[1000] = {0};
    int queue[1000], front, rear, count;
    front = rear = count = 0;

    for (int i = 0; i < prerequisitesSize; i++) {
        int course = prerequisites[i][0];
        inDegree[course]++;
    }
    for (int i = 0; i < numCourses; i++) {
        if (inDegree[i] == 0) {
            queue[rear++] = i;
        }
    }
    while (front < rear) {
        int currCourse = queue[front++];
        count++;
        for (int i = 0; i < prerequisitesSize; i++) {
            if (prerequisites[i][1] == currCourse) {
                int depCourse = prerequisites[i][0];
                inDegree[depCourse]--;

                if (inDegree[depCourse] == 0) {
                    queue[rear++] = depCourse;
                }
            }
        }
    }
```

```
    return count == numCourses;
}
```

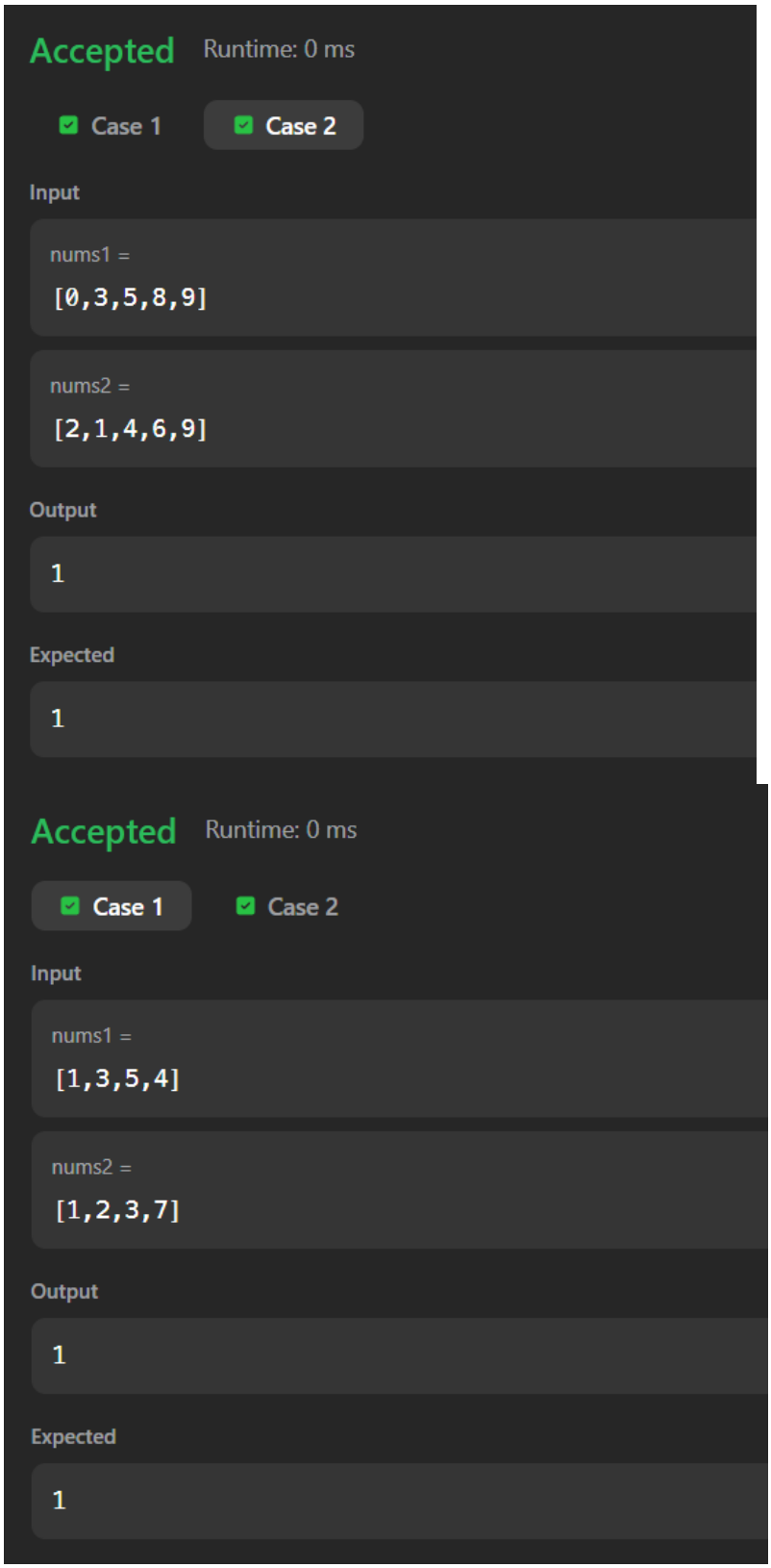
Output:



2)Leet Code exercise on Knapsack problem using dynamic Programming(801).

```
int minSwap(int* nums1, int nums1Size, int* nums2, int nums2Size) {
    int x = 0, y = 1;

    for (int i = 1; i < nums1Size; i++) {
        int a = x, b = y;
        if (nums1[i - 1] >= nums1[i] || nums2[i - 1] >= nums2[i]) {
            x = b;
            y = a + 1;
        }
        else {
            y = b + 1;

            if (nums1[i - 1] < nums2[i] &&
                nums2[i - 1] < nums1[i])
                x = (x < b) ? x : b;
                y = (y < a + 1) ? y : (a + 1);
        }
    }
    return (x < y) ? x : y;
}
```

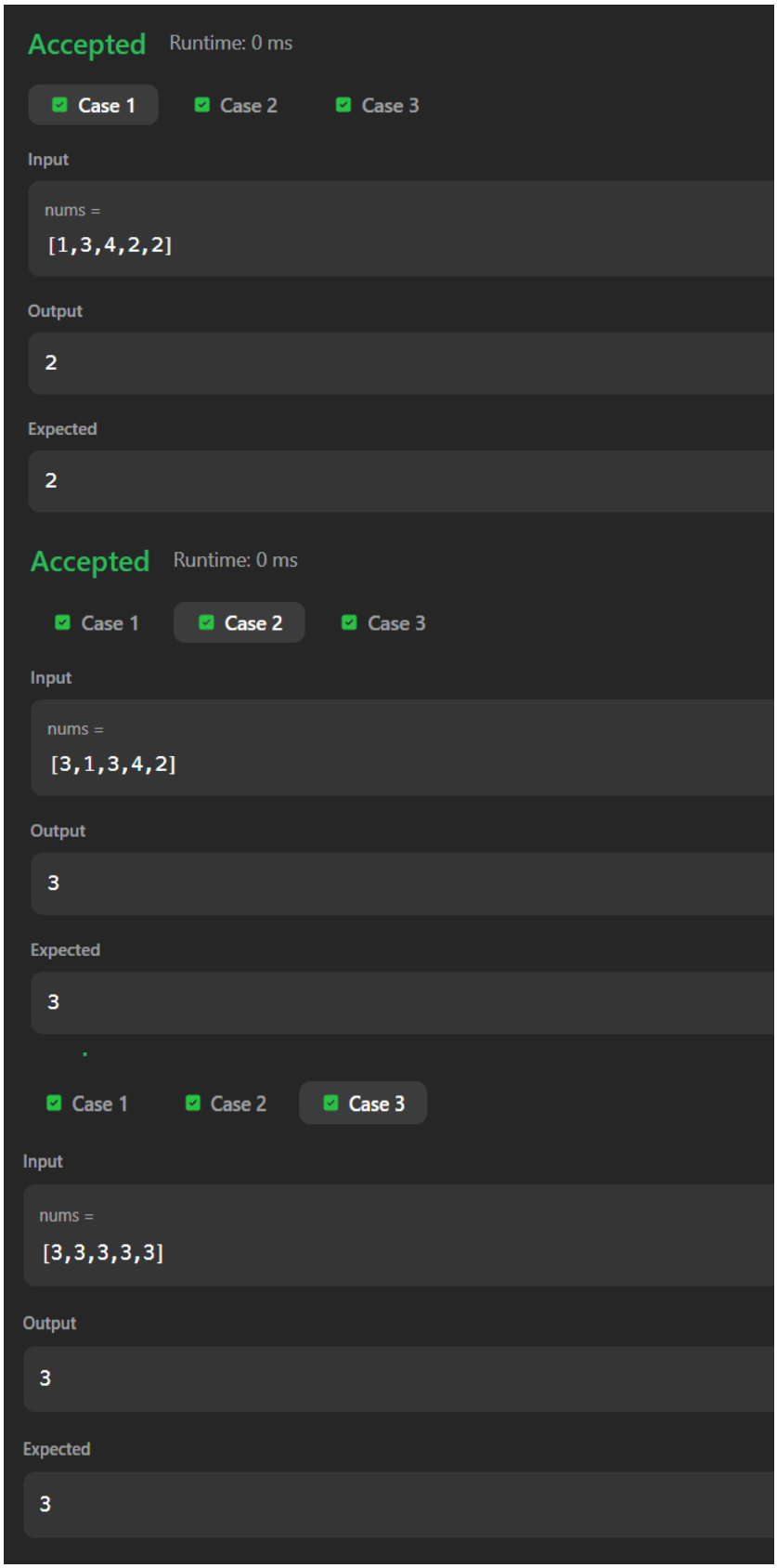
Output:


Accepted
Runtime: 0 ms
Case 1
Case 2
Input
nums1 =
[0,3,5,8,9]
nums2 =
[2,1,4,6,9]
Output
1
Expected
1
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
nums1 =
[1,3,5,4]
nums2 =
[1,2,3,7]
Output
1
Expected
1

3)Leet Code exercise on Floyd's algorithm(287).

```
int findDuplicate(int* nums, int numsSize)
{
   int slow1, fast, slow2;
   slow1=fast=slow2=0;

   while(true)
   {
      slow1=nums[slow1];
      fast=nums[nums[fast]];
      if(slow1==fast)
      {
         break;
      }
   }

   while(true)
   {
      slow1=nums[slow1];
      slow2=nums[slow2];
      if(slow1==slow2)
      {
            return slow2;
      }
   }
}
```

Output:


Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Input
nums =
[1,3,4,2,2]
Output
2
Expected
2
Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Input
nums =
[3,1,3,4,2]
Output
3
Expected
3
Case 1
Case 2
Case 3
Input
nums =
[3,3,3,3,3]
Output
3
Expected
3